

0j.py: a software tool for low complexity proteins and protein domains

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Received on February 5, 2001; Revised and accepted on March 29, 2001.



GO BACK

CLOSE FILE

Abstract

Low complexity proteins and protein domains have sequences which appear highly non-random. Over the years, these sequences have been routinely filtered out during sequence similarity searches because interest has been focused on globular proteins, and inclusion of these domains can severely skew search results. However, early work on these proteins and more recent studies of the related area of repeated protein sequences suggests that low complexity protein domains have function and therefore are in need of further investigation. Oj.py is a new tool for demarcating low complexity protein domains more accurately than has been possible to date. The paper describes Oj.py and its use in revealing proteins with repeated and poly-amino-acid peptides. Statistical methods are then employed to examine the distribution of these proteins across species, while keyword clustering is used to suggest roles performed by proteins through the use of low complexity domains.

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Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

Introduction

Low complexity proteins and protein domains have sequences which appear highly non-random, i.e. their sequences display significant regularities (Wootton, 1994). An example is collagen. Low complexity proteins are widely used by living systems to create extended/structural proteins and protein domains, as opposed to the more widely studied globular proteins. Low complexity proteins are also involved in development and in DNA binding (Wootton and Federhen, 1993). The defining work to date on low complexity in DNA and protein sequences was done by Wootton and Federhen (Wootton and Federhen, 1993) - implementing the SEG system - and Claverie and States (Claverie and States, 1993) - the XNU system. Of the two, SEG has been the more widely deployed. SEG is computationally efficient and works well at removing most non-globular domains during protein similarity searches; without it, low complexity regions can greatly skew search results. The algorithm underlying SEG is rather simplistic: if, within a sliding window of 12aa the distribution of the counts for the different amino acids is skewed, the amino acids in the window are assumed to have low complexity and are masked (i.e. replaced by X). A related, more recent algorithm is implemented in the CAST system (Promponas *et al.*, 2000). In this system, 20 poly-amino-acid peptides are created (i.e. one for each natural amino acid), and each is aligned to the target protein using the Smith Waterman algorithm, based on a standard scoring matrix but infinite gap penalties. If the highest scoring match is above a threshold, matching amino acids in the target protein corresponding to the

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

poly-aa type are masked. This process is iterated until no score is above the threshold. The advantage of the CAST algorithm is that it only masks those amino acids that found to be low complexity, rather than the entire regions containing the low complexity amino acids. In general, the approach adopted by SEG and CAST can be characterised as “low complexity through amino acid composition bias”. There are two problem with this approach. While both are able to identify low complexity regions involving numerous repetitions of the same amino acid, SEG (and to a lesser extent CAST) may overlook low complexity regions which are due to repeats. More abstractly, a sequence can have biased composition, but not be low complexity, e.g. a randomly generated sequence where one type of amino acid has much higher probability than the others (say $\text{Pr}(Q)=0.33$). Finally, Meyer and Tollet describe an experiment where a pattern matching program was employed, once for each amino acid type, to search for poly amino acid peptides of at least 4aa ([Meyer and Tollet Jr, 2001](#)).

Much work has also been done on repeats in protein sequences, e.g. ([Heringa, 1998](#)), ([Edward M. Marcotte *et al.*, 1998](#)). While some repeats can be considered as being low complexity peptides, others are very large, e.g. the numerous Kringle domains in APOA.HUMAN, each of which is approximately 110aa in length and of high complexity ([Martin, 1999a](#)). Repeated EGF domains ([Martin, 1999b](#)) are another example. Though extensively repeated, such regions are essentially globular and are stabilized with disulphide bonds. They therefore cannot be considered as low complexity regions.

Abstract

Introduction

Measuring Protein...

Algorithmics

Experimenting with...

Species...

Exploring Low...

Acknowledgements

References



GO BACK

CLOSE FILE

With this in mind, a new software tool has been created, called *0j.py*, which allows investigators to more accurately demarcate low complexity proteins and protein domains than has been possible to date. Armed with this knowledge, investigators will be able to discover the “meaning” of low complexity regions (i.e. their structure and function). This paper presents details both of *0j.py* and of an initial investigation into the meaning of low complexity proteins using the new tool.

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

Measuring Protein Complexity using Regular Expressions

0j.py measures levels of protein sequence complexity by constructing *regular expressions* which encode the sequences. (Prosite patterns are regular expressions, although the regular expressions created by 0j.py owe more to the patterns used by the UNIX utility, “sed”.) The significance of the use of regular expressions is that if low complexity regions are being encoded, the resulting regular expressions are compressed compared to the original sequences, whereas regular expressions constructed from high complexity protein sequences (e.g. globular proteins such as enzymes) are just the input sequences; no compression is possible. The compressibility of different sequences can therefore be a metric for the differing levels of sequence complexity. This approach is similar to that used by Allison et al ([Allison et al., 2000](#))), except their notion of compressibility is based on variations of the Lempel Ziv data compression algorithm, implemented as a hidden markov model and applied to DNA.

The regular expressions created by 0j.py are based on the following encoding of the amino acids which make up a protein: Assume that a “token” can either be a letter (representing a natural amino acid) or a number (representing a dictionary entry). Further assume that each token has a superscript representing the number of tandem occurrences; if there is only a single occurrence, the superscript (just 1) can be ignored. With this in mind, the following sequence (taken slightly modified from the tail of H1 Histone - H11L_CHICK):

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

AKSPAKAKAVKPKAAKPKAAKPKAAKAKKAAAKKK

might be encoded as: $1SP1^2AVK2^3AKA^3K^3$, where the dictionary contains:

1: AK

2: PKAAK

A minimum-length encoding algorithm is part of the scoring mechanism. That is, the total length of an encoded sequence is the number of tokens in the code (in the above case 12) plus the number of tokens in the dictionary entries (7); otherwise the code would consist of just a single token with the entire sequence being placed in the dictionary. Notice that, even when these overheads are taken into account, the encoded sequence in the example is compressed (19) when compared to the original sequence, 34, giving a score of 15. Notice also that tandem repeats attract a bonus compared to almost-tandem repeats (i.e. repeats separated by a small number of amino acids) because each almost-tandem repeat must be represented by a dictionary index which is saved through use of the superscript. Finally, to ensure that encodings are as simple as possible an *occam tax* of 1 is deducted for each substring-type, after the first, that is included in an encoding. The final score for the above example is therefore 12.

Implicit in the encoding scheme is a distinction between *repeated peptides* and *poly-amino-acid*, or *poly-aa*, peptides. The difference between the two types is simply that poly-aa peptides involve tandem repeats of a single amino acid, e.g. AAA, while repeated peptides can be tandem or almost-tandem repetitions of different amino acids, e.g. AK in AKSPAKAK.

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

Algorithmics

Constructing regular expressions from protein sequences involves a two phase process: first finding all repeats and poly-aa peptides in the input sequence up to a given maximum length and then finding an arrangement of the repeats and poly-aa peptides which maximizes the compression score.

In the first phase, all repeats and poly-aa peptides are recorded for a given base-length, say $2aa$. If two repeated peptides of length N overlap across $N - 1$ amino acids, say PKA and KAA, then it is possible to infer repeats of length $N + 1$, in this case PKAA. By only looking at the those suffixes of length $N - 1$ that are already present in the set of peptides of length N , it is possible to efficiently iterate the process until either the maximum peptide length has been reached or until no repeated or poly-aa peptides remain in the set. (To continue the iteration there must clearly be at least two instances of repeated peptides, but there need only be a single instance of a poly-aa peptide.)

The iteration process returns all possible repeated and poly-aa peptides, from the base-length up to the maximum length. Within this list, however, peptide instances may overlap, e.g. in the peptide AAAA there are three overlapping instances of AA. At the other extreme, the instances of a repeated peptide may be so far apart they that they must be considered as separate instances rather than as part of the same repeat. A second pass across the repeated and poly-aa peptides therefore completes the first phase of the application. In this pass, only non-overlapping repeats are retained, and repeats are split if adjacent instances of a particular repeated peptide are separated by more than the length of the

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

peptide, so, for example, the gap between repeated tripeptides can be no more than 3aa. Separate base-lengths may be set by the user for repeated and poly-aa peptides; the user can also set the maximum peptide length. No further controls are required by the application.

Out of the first phase come lists of substrings which represent particular repeated or poly-aa peptides. Because the same peptide may appear in a number of lists due to the original list for that substring having been split, and because there is no distinction in the algorithm between repeated and poly-aa peptides, from this point the discussion will refer to the substrings representing repeated or poly-aa peptides as *tiles* and the lists as *tile-lists*. Tile-lists are scored using the minimum-length encoding assumptions discussed above. This problem to be solved in the second phase therefore is to combine tile-lists so as to maximize the compression score, taking into account the occam-tax mentioned earlier, which is deducted for each tile-list after the first that is added to the combination (or *tiling*). Of course, including one tile-list in a final combination may wholly or partly occlude another. For example, the encoding returned for the above example uses PKAAK; it could have equally used KPKAA, but as these are mutually exclusive one must be chosen over the other. The problem, which can be seen to fall into the class of NP-complete problems by analogy with the set covering problem, has been addressed by this author in another context - a system, called Neweyes, for aligning sequences in the presence of transpositions (Wise, 1995). In the earlier application a greedy algorithm was devised, but it quickly became apparent that in this application a purely greedy approach

Abstract

Introduction

Measuring Protein...

Algorithmics

Experimenting with...

Species...

Exploring Low...

Acknowledgements

References



GO BACK

CLOSE FILE

would miss significantly better solutions. For that reason, a combination of techniques has been employed.

In the first pass, the list of tile-lists is traversed to see if there are any groups of tile-lists, or *patches*, which do not overlap other patches. In the example above, the tile-list for AK overlaps a shorter tile-list for KA, but these do not overlap the other tile-lists and therefore form one patch, with the remainder of the tile-lists forming a second patch. Patches can be tiled separately.

In the next step, starting with tile-lists that have been ordered by decreasing score, pairs of tile-lists are combined and the results are scored. This continues until a large set of top scoring combinations stabilizes. To speed-up each comparison, a prior test is carried out modelling the tiles as bit-sets, and the process of laying down tile-lists as the subtraction of bit-sets to see if what remains would represent viable tiles.

Aided by the bit-set pre-tests, the top-scoring pairs from the all-against-all comparisons are passed to the third step, in which individual pairs of tile-lists are extended by the addition of further tile-lists until no further extensions are possible. This is done in a greedy fashion, so that at each step in the iteration for a given starting pair of tile-lists, the tile-list whose addition produces the best score is accepted. The product from this step is a set of tilings (i.e. combinations of tile-lists); the best of these is the result for that patch. The final result is the combination of the results for the different patches.

As will be seen in the results from the experiments described below, the majority of proteins have either no low complexity domains, or else the low

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

complexity domains are limited in scope and non-overlapping (and hence constitute different patches). For these proteins, processing will be completed by the end of the first stage, at which point the set of tile-lists will be empty or perhaps contain a single tile-list. Of the remaining proteins, many with moderate amounts of low-complexity peptides are resolved after the pairwise comparison stage, and only the small minority of proteins with significant low complexity regions pass through all the stages outlined above.

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

Experimenting with 0j.py

As a first experiment it was hypothesized that repeated peptides may have a different biological meaning to poly-aa stutters. To examine this proposition, 0j.py was run three times: once with the minimum repeat length set to 2aa, but with poly-aa stutters disallowed (by setting an extreme minimum poly-aa length), a second time with the minimum stutter length set to 3aa, but with repeats disallowed and finally with the minimum repeat length set to 2aa and the the minimum poly-aa length set to 3aa. Each experiment was run against SwissProt, as at January 4, 2001, which held 91,131 protein records. For each protein in an experiment, two scores are returned: an overall score as described above, and the score for the highest scoring contiguous segment, or *HSCS*. A HSCS may be longer than a single patch because, although patches - by definition - do not overlap, two adjacent patches may be joined if they are sufficiently close based on the repeats-splitting rule described earlier. It is believed that HSCS scores are more useful for evaluating low complexity protein domains because the scores are not correlated with protein length, as evidenced by the correlation coefficient (r^2) values in [Table 1](#).

It is also hypothesized that use of HSCS scores better models low complexity domains within otherwise high complexity proteins, e.g. the tail of the H1 histone, or the yeast SUP35 (Omnipotent suppressor protein) particle within ERF2_YEAST. However, a small, but significant number of proteins such as MUC2_HUMAN and SPD2_NEPCL have both repeated and poly-aa peptides. This can cause problems because if one repeat-type is being ignored, peptides

Abstract

Introduction

Measuring Protein...

Algorithmics

Experimenting with...

Species...

Exploring Low...

Acknowledgements

References



GO BACK

CLOSE FILE

Table 1. r^2 score vs length for poly-aa and repeated peptides.

	HSCS Score	Total Score
Poly-aa	0.084	0.231
Repeats	0.020	0.013
Both	0.023	0.116

Table 2. Statistics on HSCS and overall scores for poly-aa and repeated peptides.

	Repeats Only		Poly-aa Only	
	HSCS	Overall	HSCS	Overall
Mean	1.51	1.77	2.00	3.59
Std-Dev	6.22	10.28	2.32	5.83
Mode	1	1	2	0
95%	3.83	3.72	4.15	10.28
Max	708	2168	159	520

of this type may constitute sufficiently long gaps so that patches are broken up and HSCS scores are artificially lowered. The recommendation therefore is that when both low complexity peptide types are being measured, HSCS scoring should be used, but when one or other species is being measured, total scores should be use.

With this in mind, the basic statistics of the distributions of the total and HSCS scores for repeats-only and poly-aa-only scans with 0j.py are outlined in [Table 2](#).

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

A number of inferences may be drawn from the above statistics and from those for the third experiment (which allows both repeated and poly-aa peptides). The most obvious fact is that the vast majority of proteins have minimal low complexity content: 27% have no poly-aa peptides of at least 3aa while 28% have no repeats sufficient to score at least 1 (either 3 repeats of a dipeptide or two repeats of a tripeptide). Of these, almost 50% (i.e 13% of the entire database) have neither repeated nor poly-aa peptides sufficient to score at least 1. Of those proteins that have low complexity regions, scores of 1, 2 or 3 predominate pointing again to the fact that the majority of proteins are high complexity. However, the distributions of the scores also have very long tails showing that a minority of proteins possess very significant areas of low complexity.

Another statistic of interest is the degree of commonality between the ranked lists returned for the repeats-only and poly-aa-only experiments (based on the overall scores). [Table 3](#) records the number of shared entries found within the the N highest scoring entries taken from the respective lists.

It can be seen that the bulk of very low complexity proteins have a preference for either repeated or poly-aa peptides, though a significant number contain both types.

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

Table 3. Counts of proteins found both in the top N of the poly-aa list and the repeated peptide list.

Top N	Shared	% of N
100	6	6
200	28	14
300	53	18
400	80	20
500	104	21
600	130	22
700	151	22
800	181	23
900	216	24
1000	252	25

[Abstract](#)
[Introduction](#)
[Measuring Protein ...](#)
[Algorithmics](#)
[Experimenting with ...](#)
[Species ...](#)
[Exploring Low ...](#)
[Acknowledgements](#)
[References](#)



GO BACK

CLOSE FILE

Species Distribution of Low Complexity Peptides

Looking at the 2,000 top scoring proteins from the repeats-only poly-aa-only experiments, one can also examine the range of species possessing these unusual peptides: 389 species are represented in the list from the repeats-only experiment, while 283 species are found in the poly-aa-only list. However, using a binomial distribution statistic and assuming a null hypothesis of an even distribution of low complexity regions across species, it is clear from [Table 4](#) that there is a significant skew in the distribution of both repeated and poly-aa peptides across species. The top half of the table records species that are over-represented with probability no greater than 10^{-6} . The lower half of the table are those species that are significantly under-represented at the same probability level. The species shown *italics* are significantly over-represented (as indicated by their probability values), but as they have 15 or fewer records in SwissProt there must be some doubt about the representativeness of the few proteins found in SwissProt versus the protein complements of the organisms as a whole.

Finally, two species, *Haemophilus influenzae* and *Methanococcus jannaschii* appear under-represented in the repeats-only list (in the former case the level is marginal: 0.0027), but are absent from the poly-aa-only list and this is statistically significant (probabilities of 2.0e-09 and 5.7e-07 respectively).

In contrast, 10 random collections of the same size covered 595.5 species on average and had average under-representation and over-representation probabilities of 0.014 and 0.00083.

[Abstract](#)[Introduction](#)[Measuring Protein ...](#)[Algorithmics](#)[Experimenting with ...](#)[Species ...](#)[Exploring Low ...](#)[Acknowledgements](#)[References](#)[GO BACK](#)[CLOSE FILE](#)

Table 4. Distribution of species among 2,000 top scoring proteins, with probabilities based on a binomial distribution statistic.

Repeats Only	Poly-aa Only
<i>Drosophila melanogaster</i> (6.6e-98)	<i>Drosophila melanogaster</i> (2.2e-253)
<i>Plasmodium falciparum</i> (2.0e-44)	<i>Homo sapiens</i> (6.6e-98)
<i>Homo sapiens</i> (2.3e-19)	<i>Saccharomyces cerevisiae</i> (1.3e-57)
<i>Dictyostelium discoideum</i> (2.2e-16)	<i>Mus musculus</i> (3.8e-28)
<i>Chironomus tentans</i> (9.3e-11)	<i>Dictyostelium discoideum</i> (1.2e-23)
<i>Triticum aestivum</i> 5.4e-09	<i>Candida albicans</i> (7.4e-09)
<i>Plasmodium cynomolgi</i> (6.5e-09)	<i>Plasmodium falciparum</i> (8.8e-08)
<i>Locusta migratoria</i> (7.3e-09)	<i>Drosophila virilis</i> (1.4e-08)
<i>Plasmodium knowlesi</i> (1.6e-08)	Hepatitis C virus (1.1e-08)
<i>Tenebrio molitor</i> (2.3e-08)	<i>Gallus gallus</i> (5.7e-07)
Epstein-barr virus (3.1e-07)	<i>Rattus norvegicus</i> 5.7e-07)
<i>Plasmodium berghei</i> (3.7e-07)	<i>Dengue virus type 2</i> (2.2e-06)
<i>Bombyx mori</i> (5.2e-07)	
Herpes simplex virus type 1 (1.6e-06)	
<i>Strongylocentrotus purpuratus</i> (1.5e-06)	
<i>Escherichia coli</i> (1.2e-15)	<i>Escherichia coli</i> (1.5e-23)
<i>Bacillus subtilis</i> (2e-09)	<i>Bacillus subtilis</i> (2e-09)
<i>Methanococcus jannaschii</i> (5.7e-07)	

From the above figures it can be seen that certain species appear to make extensive use of both poly-aa and repeated peptides (*D. melanogaster*, *H. sapiens*, *D. discoideum*), others predominantly use one form or the other

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

(*M. musculus*, *S. cerevisiae*) and some appear to avoid low complexity protein domains entirely. It would therefore appear that eukaryotes are more likely to have proteins with low complexity domains, and eubacteria are unlikely to have them. Viruses and eubacteria which have proteins with low complexity domains have perhaps acquired them through gene-transfer from eukaryote hosts; there are 10 instances of Bacteriophages in the repeats-only and poly-aa only lists (1 reference for each of 10 species), 9 are not significant because the number of protein records are too small. The last, T4, is a marginally significant under-representation (Probability 0.0025) in the repeats-only list, but is absent from the poly-aa-only list. However, other bacteriophages with a reasonable number of records in SwissProt, such as Mycobacteriophage L5, Mycobacteriophage D29, Bacteriophage P22 and Bacteriophage lambda are absent from both lists. Finally, this finding is, in part, similar to a conclusion drawn by [Edward M. Marcotte *et al.* \(1998\)](#), that protein repeats are more prevalent in eukaryotic proteins compared with prokaryotic proteins.

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

Exploring Low Complexity Proteins using PAA

Another way to explore the connections between proteins is to use the Protein Annotators' Assistant. The Protein Annotators' Assistant (or PAA), <http://www.ebi.ac.uk/paa/>, is a software application which uses stemming and keyword clustering to find keywords and phrases that are common to groups of proteins from a list submitted by the user. PAA is typically used to propose functions for proteins which have been the subject of a database similarity search. In that situation the hits returned by the particular search engine are sent to PAA, the assumption being that if these proteins have a particular keyword/phrase in common, and they are related to a query protein, then the keyword/phrase might also describe the query protein.

In the present application the input data for PAA are subsets of the list of proteins returned by Oj.py. The version of PAA used in this study, 0.93, is essentially the same as that described in (Wise, 2000a) and (Wise, 2000b), except the keywords and phrases are scored by their significance within the input set versus their significance across SwissProt as a whole, based on a binomial distribution statistic. It is also now possible to get an estimate for the significance of the entire set of keywords returned by PAA (and thereby the relatedness of the input set of proteins), via the average log probability score.

The analysis summarized in earlier tables suggests the existence of three types of proteins containing significant low complexity regions: those with significant repeated peptides, but with far less or no poly-aa peptides; those with significant poly-aa peptides but far less or no repeated peptides, and those

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

Table 5. Keywords and phrases corresponding to the top 252 proteins unique to either the poly-aa list, the repeated peptide list or common to both lists.

Type	Keywords/Phrases (Probability)
Repeats Only	malaria (7.7e-71), signal (7.3e-51), proline (3.3e-32), antigen (1.3e-35), collagen alpha (4.2e-34), surface (9.8e-29), ice nucleation (1.2e-18), extracellular matrix (7.3e-18), proteoglycan (6.4e-17)
Poly-aa Only	nuclear protein (5.4e-211), dna binding (2.9e-107), transcription (2.9e-107), homeobox (1.4e-28), development (4e-26), embryo (2.6e-25), zinc finger (5.3e-22), brain (2.6e-18), phosphorylation (1.2e-15)
Common to both lists	nuclear protein (1.2e-38), transcription (3.8e-28), rna recognition motif (9.3e-26), intermediate filament (1.3e-21), dna binding (2.3e-19), coiled coil (1.1e-16), rna binding (6e-15)

proteins that have significant amounts of both types. To further examine the ways in which the proteins in these sets differ, the 252 proteins common to both lists were submitted to PAA. A similar number of proteins from the repeats-only list and from the poly-aa-only list were also submitted, excluding any shared between the lists. [Table 5](#) lists some of the keywords/phrases that resulted.

As an indication of the quality of the respective lists of keywords/phrases, the average log probability values are 7.9, 6.7 and 6.0, versus an average of 2.7 for five groups of 200 randomly selected proteins. Note that constellations of keywords often appear together, e.g. many of the proteins mentioning “malaria” also mention “sporozoite”, so the above list has included the higher scoring or more general terms.

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

From the keyword lists in [Table 5](#) it seems that repeated peptides have a different meaning than poly-aa peptides; the former are extra-cellular or appear on membrane surfaces and are involved with connective tissue, while the latter are nuclear and involved with DNA binding and regulation. Secondly, while there is some overlap, proteins containing significant amounts of both species fill a third niche, e.g. involving intermediate filaments.

It is possible to drill down into these lists, though in a limited way, e.g. to see whether different poly-aa peptides have different meanings. The problem is that proteins with significant levels of poly-aa peptides are relatively rare, so when this number is further subdivided by the specific amino acid that is repeated, numbers can be too small to yield significant keywords shared by more than a tiny number of proteins. An additional caveat is that poly-aa repeats involving one amino acid rarely appear alone, but often are associated with poly-aa peptides involving other amino acids. For this reason the top 2,000 highest scoring proteins from the poly-aa list were used. [Table 6](#) lists keywords associated with poly-aa peptides involving specific amino acids, together with the number of proteins which have poly-aa peptides involving that amino acid.

Examples of proteins containing significant poly-aa peptides are rare among the entries found in PDB; see Table 3 of ([Meyer and Tollet Jr, 2001](#)). However, some idea of which secondary structure class a poly-aa peptide might belong to may be obtained using the various protein secondary-structure-prediction programs which are available. Specifically, for each amino acid that is found in poly-aa peptides among the top 2,000 proteins in the poly-aa-only list,

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

a poly-aa sequence of length 40aa was created. In addition, if the IUPAC code might otherwise be misconstrued as a nucleotide (and the sequence therefore rejected by some of the secondary-structure prediction servers), a P was attached to each end of the sequence. This sequence was submitted to the following secondary-structure-prediction programs: PHD (Rost, 1996), PSIPRED (Jones, 1999), PSSP <http://imtech.ernet.in/raghava/pssp/>, SAM-T99 (Karplus *et al.*, 1998) and SSPro (Baldi *et al.*, 1999). These cover a number of different prediction techniques and are launched via a single server, PP Meta Server http://maple.bioc.columbia.edu/pp/submit_meta.html. Where there was any dissent for a particular poly-aa sequence, the sequence was also sent to a alternative set of secondary structure prediction servers: DPM (Deleage and Roux, 1987), DSC (King and Sternberg, 1996), GOR IV (Garnier *et al.*, 1996), HNN (Guermeur, 1997), Predator (Frishman and Argos, 1996), Simpa96 (Levin, 1997) and SOPM (C. Geourjon, 1994). These are launched through the NPS@ server, <http://pbil.ibcp.fr/>. The third column of Table 6 records the majority verdict about the poly-aa sequence: H (alpha helix), E (beta sheet) or C (random coil).

Not listed in the above are poly-aa peptides for which too few examples exist in the poly-aa-only list: poly-C (2), poly-F (4), poly-I (4) and poly-Y (3), or for which no significant keyword associations could be found: poly-M (10) and poly-V (22).

From the above table it can be seen that many of the poly-aa peptides are significantly involved in DNA binding and reside in the nucleus (particularly

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

poly-A, poly-Q, poly-G and poly-S). Development is another prominent keyword, (poly-Q, poly-S and poly-H), as are the related concepts homeobox (poly-A, poly-G and poly-Q) and embryo. Poly-aa sequences are found in the brain/CNS (poly-P, poly-H, poly-A, poly-G), kidneys (poly-P), neurons (poly-E), muscles (poly-K, poly-P) and sperm (poly-R). Poly-T is heavily presented in mucus and Drosophila salivary glue, and appears to be one of the few poly-aa that forms beta sheets. While low complexity peptides are generally found in eukaryotes, poly-L is associated with capsid proteins; poly-L also has the most significant probability score for the keyword glycoprotein. (Glycoprotein is also associated with poly-T and poly-S; the former is understandable in view of the extra-cellular role of the proteins, while the latter is largely due to the appearance of poly-S in proteins associated with the keyword signal - probability $5.7e-07$; of the 83 proteins associated with keyword glycoprotein and 96 associated with signal, 60 are associated with both keywords).

This is clearly just a first exploration of the universe of low-complexity protein domains using the first version of 0j.py, but from these results it is clear that low-complexity peptides have significant meaning that has hitherto not been adequately explored. 0j.py, and similar software tools, will greatly assist in determining the character of the members of this universe.

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

Table 6. Keywords and phrases associated with proteins containing poly-aa peptides for each type of amino acid, and the consensus secondary structure.

AA	#	2 nd	Keywords/Phrase (Probability.)
A	314	H	nuclear protein (1.1e-224), dna binding (5e-149), transcription regulation (4.7e-117), homeobox (4.4e-44), zinc finger (5.9e-26), nervous (6.7e-15), development (1.4e-15), embryo (4.1e-14), forkhead (4e-13), cns (2.4e-11)
D	107	C	nuclear (1e-28), dna binding (6.1e-16), eukaryotic topoisomerase (4.7e-12), transcription (7.7e-12), phosphorylation (3.6e-08), hmg box (1.1e-07)
E	276	H	nuclear (7.3e-51), phosphorylation (1.2e-38), excitable (5.9e-25), neurofilament (1.4e-22), coiled coil (2.5e-22), filament (1.5e-20), ionic channel (3.5e-18), intermediate filament (1.8e-16), neuronal (1.8e-15)
G	293	C	nuclear (1.5e-160), transcription (1.9e-72), dna binding (1.3e-57), rna recognition motif (6.3e-40), microfibrillar keratin (1.8e-33), homeobox (3e-31), rna binding (1.8e-30), intermediate filament (9.7e-30), cytoskeletal (1.5e-29), coil (7.8e-27), ribonucleoprotein (1.5e-18), circadian (3.4e-16), brain (1.9e-12)
H	91	C	nuclear protein (1.3e-42), dna binding (5.6e-29), development (1.9e-22), transcription regulation (3.7e-20), brain (7e-18), calcium channel (3.5e-17), zinc finger (4.2e-16), homeobox (2.4e-15), neuron (2.2e-14)
K	68	H ¹	myofibril (5.7e-12), muscle contraction (1.2e-11), nuclear protein (2.1e-10), atpase (7.8e-06)
L	102	H	capsid protein c (4.6e-35), glycoprotein (7.4e-34), transmembrane (1.5e-23), rna directed rna polymerase (7.1e-21), lipoprotein (2.1e-15), atp binding (1.7e-07)
			Continued...

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

Table 6 Continued...

N	131	C	nuclear (2.8e-20), prestalk (4.7e-12), development (4.7e-12), dna binding (9.1e-11), threonine protein kinase (9.8e-10), kinase (1.6e-09), camp (4.8e-09)
P	274	C	nuclear protein (6.6e-98), transcription (1.2e-38), dna binding (3.6e-33), brain (8.2e-24), oncogene (1.7e-19), neuron (8.4e-18), rna binding (5e-15), kidney (6.2e-14), development (4e-11), muscle (2.8e-10)
Q	337	H	nuclear protein (8.1e-239), transcription (6.1e-138), dna binding (2.8e-127), segmentation (3e-40), development (6.4e-39), homeobox (1.4e-25), embryo (6.2e-25), metal binding (1.2e-19), cns (1.7e-13)
R	86	H	sperm dna (1.5e-33), chromatin (3.3e-25), nuclear protein (2.4e-20), brain (5.2e-14), phosphorylation (4.7e-10)
S	383	C	nuclear protein (2.9e-107), dna binding (1.3e-57), transcription (1.6e-44), phosphorylation (3.8e-28), development (4e-28), zinc finger (3.2e-18), intermediate filament (2.2e-14), coiled coil (1.1e-13), glycoprotein (2.6e-12)
T	110	E ²	clathrin binding (1.7e-16), signal (1.2e-15), glycoprotein (2.6e-12), salivary glue protein (1.9e-12), alzheimer (3.2e-08), embryo (2.9e-08), mucin (1.6e-08)

Notes

*H*¹ One the initial five systems, the verdict on poly-K (poly-lysine) was split between C, H and alternating C and H. The split was also reflected among the second set of servers, but a majority opted for H.

*E*² Predictions for poly-T were affected by the problem of sequence rejection because of a possible confusion with the nucleotide Thymine. The remaining servers were split between three votes for E and two for C.

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

Acknowledgements

Dr Wise is Senior Research Fellow at Pembroke College and CCSR under a grant from Bristol-Myers Squibb.

Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

References

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[Abstract](#)

[Introduction](#)

[Measuring Protein ...](#)

[Algorithmics](#)

[Experimenting with ...](#)

[Species ...](#)

[Exploring Low ...](#)

[Acknowledgements](#)

[References](#)



[GO BACK](#)

[CLOSE FILE](#)

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Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE

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Abstract

Introduction

Measuring Protein ...

Algorithmics

Experimenting with ...

Species ...

Exploring Low ...

Acknowledgements

References



GO BACK

CLOSE FILE